LAB 4 – SQL FUNCTIONS

Conversion Functions

```
CREATE TABLE employees (
    employee_id INT PRIMARY KEY,
    first_name VARCHAR(50),
    last_name VARCHAR(50),
    email VARCHAR(100) UNIQUE,
    hire_date DATE,
    department_id INT
);

INSERT INTO employees VALUES

  (101, 'Aarav', 'Sharma', 'aarav.s@company.com', '2020-05-10', 1),
  (102, 'Rohan', 'Mehta', 'rohan.m@company.com', '2019-03-15', 2),
  (103, 'Kavya', 'Nair', 'kavya.n@company.com', '2021-07-20', 3),
  (104, 'Divya', 'Iyer', 'divya.i@company.com', '2018-11-01', 4),
  (105, 'Nikhil', 'Rao', 'nikhil.r@company.com', '2022-01-12', 2),
  (106, 'Meera', 'Joshi', 'meera.j@company.com', '2020-09-23', 3),
  (107, 'Vikram', 'Patel', 'vikram.p@company.com', '2021-04-18', 1),
  (108, 'Sneha', 'Reddy', 'sneha.r@company.com', '2019-08-30', 4),
  (109, 'Imran', 'Khan', 'imran.k@company.com', '2022-11-05', 3),
  (110, 'Tanya', 'Gupta', 'tanya.g@company.com', '2023-02-10', 2);

SELECT * FROM employees;
```

employee_id	first_name	last_name	email	hire_date	department_id
101	Aarav	Sharma	aarav.s@company.com	2020-05-10	1
102	Rohan	Mehta	rohan.m@company.com	2019-03-15	2
103	Kavya	Nair	kavya.n@company.com	2021-07-20	3
104	Divya	Iyer	divya.i@company.com	2018-11-01	4
105	Nikhil	Rao	nikhil.r@company.com	2022-01-12	2
106	Meera	Joshi	meera.j@company.com	2020-09-23	3
107	Vikram	Patel	vikram.p@company.com	2021-04-18	1
108	Sneha	Reddy	sneha.r@company.com	2019-08-30	4
109	Imran	Khan	imran.k@company.com	2022-11-05	3
110	Tanya	Gupta	tanya.g@company.com	2023-02-10	2

Implicit Data Type Conversion

```
SELECT employee_id, first_name, department_id
FROM employees
WHERE department_id > 2;
```

employee_id	first_name	department_id
103	Kavya	3
104	Divya	4
106	Meera	3
108	Sneha	4
109	Imran	3

Explicit Data Type Conversion

```
SELECT employee_id, first_name,
       CAST(hire_date AS CHAR) AS hire_date_text
```

```
FROM employees;
```

employee_id	first_name	hire_date_text
101	Aarav	2020-05-10
102	Rohan	2019-03-15
103	Kavya	2021-07-20
104	Divya	2018-11-01
105	Nikhil	2022-01-12
106	Meera	2020-09-23
107	Vikram	2021-04-18
108	Sneha	2019-08-30
109	Imran	2022-11-05
110	Tanya	2023-02-10

Date Functions

```
SELECT
    first_name,
    last_name,
    hire_date,
    CURDATE() AS today,
    YEAR(hire_date) AS year,
    MONTH(hire_date) AS month,
    DAY(hire_date) AS day,
    YEAR(CURDATE()) - YEAR(hire_date) AS approximate_years
FROM employees;
```

first_name	last_name	hire_date	today	year	month	day	approximate_years
Aarav	Sharma	2020-05-10	2026-10-02	2020	5	10	6
Rohan	Mehta	2019-03-15	2026-10-02	2019	3	15	7
Kavya	Nair	2021-07-20	2026-10-02	2021	7	20	5
Divya	Iyer	2018-11-01	2026-10-02	2018	11	1	8
Nikhil	Rao	2022-01-12	2026-10-02	2022	1	12	4
Meera	Joshi	2020-09-23	2026-10-02	2020	9	23	6
Vikram	Patel	2021-04-18	2026-10-02	2021	4	18	5
Sneha	Reddy	2019-08-30	2026-10-02	2019	8	30	7
Imran	Khan	2022-11-05	2026-10-02	2022	11	5	4
Tanya	Gupta	2023-02-10	2026-10-02	2023	2	10	3

LAB 5 – SQL OPERATORS

```
SELECT
    employee_id,
    first_name,
    department_id,

    employee_id + 10 AS id_plus_10,

    employee_id > 105 AS id_greater_than_105,

    department_id = 2 AND employee_id > 105 AS dept2_and_id_above_105,
    department_id = 2 OR department_id = 3 AS dept2_or_dept3,
    NOT department_id = 2 AS not_dept2

FROM employees;
```

employee_id	first_name	department_id	id_plus_10	id_greater_than_105	dept2_and_id_above_105	dept2_or_dept3	not_d
101	Aarav	1	111	0	0	0	1
102	Rohan	2	112	0	0	1	0
103	Kavya	3	113	0	0	1	1
104	Divya	4	114	0	0	0	1
105	Nikhil	2	115	0	0	1	0
106	Meera	3	116	1	0	1	1
107	Vikram	1	117	1	0	0	1
108	Sneha	4	118	1	0	0	1
109	Imran	3	119	1	0	1	1
110	Tanya	2	120	1	1	1	0

LAB 6 – SPECIAL OPERATORS & SET OPERATIONS

Special Operators

```
SELECT
    employee_id,
    first_name,
    department_id,
    email
FROM employees
WHERE employee_id BETWEEN 103 AND 108
   OR department_id IN (1, 3)
   OR first_name LIKE 'A%'
   OR department_id IS NULL;
```

employee_id	first_name	department_id	email
101	Aarav	1	aarav.s@company.com
103	Kavya	3	kavya.n@company.com
104	Divya	4	divya.i@company.com
105	Nikhil	2	nikhil.r@company.com
106	Meera	3	meera.j@company.com
107	Vikram	1	vikram.p@company.com
108	Sneha	4	sneha.r@company.com
109	Imran	3	imran.k@company.com

Set Operations

Union

```
SELECT first_name
FROM employees
WHERE department_id = 1

UNION

SELECT first_name
FROM employees
WHERE department_id = 3;
```

first_name
Aarav
Vikram
Kavya
Meera
Imran

Intersect

```
SELECT department_id
FROM employees
WHERE employee_id <= 105

INTERSECT

SELECT department_id
FROM employees
WHERE employee_id >= 103;
```

department_id
1
2
3
4